

Customer Journey Intelligence: An Explainable, AI-Driven Multidisciplinary Framework for Personalised Digital Marketing with Next Best Marketing Action

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Abstract—Digital marketing platforms collect behavioural data at many touchpoints, yet most analytics systems answer isolated questions such as click probability or purchase probability instead of modelling the customer journey as a connected sequence. This paper presents Customer Journey Intelligence, an explainable and multidisciplinary framework that converts historical e-commerce interaction logs into chronological, session-aware customer journeys; classifies each customer into a journey stage and behavioural segment; predicts the next likely event and the purchase propensity; and recommends a Next Best Marketing Action (NBMA) for marketers. The action set explicitly includes Wait and Suppress Marketing, so that taking no action is treated as a valid and deliberate decision. The framework combines recency-frequency-monetary features, K-Means segmentation, a Markov-chain baseline, classical machine learning (Logistic Regression, Random Forest, XGBoost) and a sequence-aware model (LSTM/GRU), with SHAP-based and rule-based explanations feeding a transparent decision engine and a personalisation engine. The system is delivered through a web application with funnel, drop-off, segment and product dashboards. Evaluation on the open RetailRocket dataset uses chronological train and test splits to avoid leakage, and an offline simulation compares generic, segment-based and journey-intelligence strategies. We position the work against recent literature, identify the gap that no reviewed system combines these capabilities, and define four research questions with a reproducible evaluation protocol. This paper reports the framework design and evaluation protocol; experimental results are marked as pending.

Index Terms—customer journey, digital marketing, next best action, explainable AI, customer segmentation, purchase prediction, sequence modelling, personalisation

I. INTRODUCTION

In e-commerce and retail, customers generate behavioural data at many touchpoints, including advertisements, website visits, product searches, product views, carts and post-purchase

interactions, before a purchase decision is made. The role of digital marketing has shifted from one-time, mass-broadcast campaigns toward continuous, data-driven engagement that adapts to how customers actually behave. Comprehensive reviews confirm that predictive analytics and machine learning are increasingly central to this shift toward adaptive, customer-responsive strategy [1].

Despite this progress, most existing analytics systems still answer narrow, isolated questions, such as click-through probability, purchase probability or which single product to recommend, rather than modelling the customer's journey as a connected sequence. Recent journey-optimisation research shows that integrating behavioural signals into a single adaptive framework can improve predictive accuracy over isolated, rule-based models [2]. In day-to-day practice, marketers commonly rely on funnel dashboards, RFM-style customer segments and big-data-driven targeting rules to personalise campaigns [3], [4]. These practices are effective for broad targeting but generally stop short of two things: building an explicit, chronological journey for each customer, and explaining why a specific marketing action is recommended at a specific moment. They also rarely treat “do nothing” as a first-class decision, even though contacting a customer who has just purchased, or who is not yet ready, wastes budget and can harm the customer experience.

This paper addresses these shortcomings with a Customer Journey Intelligence framework. The framework is multidisciplinary: it draws on marketing (journey stages, segmentation, retargeting, cross-sell), data engineering (event standardisation and sessionisation), machine learning (clustering, sequence and classification models), explainable AI (Shapley and rule-

based reasoning) and software engineering (a deployable analytics platform). The main contributions are as follows.

- 1) A session-aware *Customer Journey Builder* that turns raw interaction logs into chronological customer profiles, journey stages and engineered behavioural features.
- 2) A combined analytical layer for journey-stage classification, behavioural/RFM/ML-based segmentation, next-event prediction and purchase-propensity prediction, with a systematic comparison of a Markov-chain baseline, classical machine learning and a sequence-aware model.
- 3) An *explainable Next Best Marketing Action engine* whose action set formally includes Wait and Suppress Marketing, with feature-based reasoning attached to every recommendation.
- 4) Marketer-facing journey and marketing analytics dashboards (funnel, drop-off, journey flow, segment and product analytics).
- 5) A reproducible offline evaluation protocol built around four research questions, including a comparison of generic, segment-based and journey-intelligence marketing strategies.

The work is aligned with United Nations Sustainable Development Goals SDG 8 (Decent Work and Economic Growth), by helping businesses allocate marketing effort more efficiently, and SDG 9 (Industry, Innovation and Infrastructure), by providing an open and deployable AI analytics infrastructure for digital businesses.

The remainder of the paper is organised as follows. Section II reviews related work, Section III identifies the research gap, Section IV formulates the problem, Section V states the scope, Section VI describes the dataset, Section VII presents the methodology, Section VIII describes the architecture and implementation, Section IX defines the experimental design, Section X discusses the results to be reported, Section XI covers ethics and societal impact, Section XII discusses limitations, and Section XIII concludes.

II. RELATED WORK

A. Segmentation and Strategic Marketing

A structured review of segmentation methods for e-commerce targeting finds that RFM-based representation combined with clustering, such as K-Means, is the most frequently used combination in the reviewed literature, mainly because of its scalability and interpretability. The same review notes that there is no consensus on the metrics used to compare segmentation quality, so no method can be called objectively best, and it identifies open questions around deep-learning-based segmentation [3]. At the strategy level, firms that integrate big-data analytics into a dynamic capabilities model make more adaptive, evidence-driven marketing decisions than firms relying on static segmentation alone [4]. Machine-learning analysis of large-scale behavioural and social data has also been used to reveal decision-making patterns that improve targeted engagement; in the hospitality sector, Support Vector

Regression, topic modelling and K-Means were applied to hotel reviews to predict satisfaction and segment travellers [5].

B. Explainable and Dynamic Personalisation

An explainable framework that integrates SHAP with quality function deployment (QFD) for the transition from Customer Relationship Management (CRM) to Customer Journey Management (CJM) shows that tree-based models paired with Shapley explanations can identify the key behavioural determinants of engagement and purchase and translate them into operational, budget-constrained plans [6]. Counterfactual analysis of customer-behaviour transitions, rather than static segment membership, improves the granularity of personalised marketing decisions; in that study K-Means with Euclidean distance and Logistic Regression were reported as the strongest performers [7]. Combining segmentation with machine learning also improves personalised-marketing outcomes in financial services, where K-Means obtained the highest silhouette score among the clustering methods compared and an A/B test indicated improved marketing success metrics [8].

C. Purchase and Intent Prediction

Tree-based gradient boosting has been shown to outperform traditional classifiers for purchase-behaviour prediction on real e-commerce data. XGBoost with multi-feature fusion achieved an accuracy of 0.9761, precision of 0.9763, recall of 0.9761 and ROC-AUC of 0.9768, outperforming KNN, SVM, Random Forest and a back-propagation neural network [9]. For anonymous customers, a clickstream-based approach using customer-behaviour-tendency and product-popularity features predicted purchase intention from session-level browsing behaviour, with a best F1 of 0.9031 at a two-day sliding window [10]. These results come from different datasets and settings and are therefore not directly comparable with each other or with the present study.

D. Personalisation and Engagement

Consumer-behaviour research using structural equation modelling shows that trust and perceived usefulness significantly drive engagement with AI-driven personalisation in social-media marketing, whereas privacy concern did not have a significant effect and AI personalisation itself had no significant direct effect on engagement [11]. This motivates presenting personalised actions together with clear reasoning rather than as black-box outputs.

E. Summary of Reviewed Studies

Table I consolidates the datasets, methods and key findings of the reviewed studies. Two observations shape our design. First, only a small number of the studies report complete classification tables, and several are reviews or qualitative studies for which accuracy-type metrics do not apply. Second, the empirical studies target one question at a time (segmentation, explanation, prediction or allocation), not the full path from raw events to an explained action.

TABLE I
SUMMARY OF REVIEWED LITERATURE

Ref.	Dataset / data source	Algorithms or approach	Best model and key finding
[1]	Not applicable (review)	Synthesis of AI/ML studies in digital marketing	Review; no own metrics. Predictive analytics and ML central to adaptive marketing
[2]	Real transaction and behavioural data from a Taiwanese financial institution (80:20 split, 10-fold CV)	LDA, LightGBM, CNN, LSTM, CNN-LSTM, and Binary Differential Evolution for channel allocation	LDA with a deep-learning ensemble; about 15% relative accuracy improvement over traditional ML (relative, not absolute)
[3]	Review of many e-commerce segmentation studies	RFM representation with K-Means	K-Means most frequently used, not proven best; no consensus on segmentation metrics
[4]	Qualitative, firm-level study	Dynamic-capabilities model (conceptual)	No benchmarked accuracy metric; integrated big data supports adaptive decisions
[5]	1,724 TripAdvisor reviews of 30 four/five-star Mecca hotels	SVR (SMO), LDA, K-Means	SVR gave the best regression fit; classification metrics do not apply
[6]	412,580 users and 156 source/medium pairs (Mar–Nov 2020) plus 3,492,632 product-review records (Jan 2021–Aug 2022)	Clustering, LDA, LightGBM with SHAP, benchmarked against LSTM and CNN; MILP for budget allocation	LDA-LightGBM: comparable accuracy to LSTM/CNN with far faster training
[7]	Dataset not named in the open abstract	K-Means (several distance metrics), classifiers, counterfactual analysis	Euclidean distance and Logistic Regression reported as strongest
[8]	Proprietary financial-services campaign data	Two-stage neural network; K-Means, Agglomerative, Mean-Shift	K-Means silhouette 0.62 vs. 0.61 and 0.57; A/B test showed over 100 basis-point improvement
[9]	135,968 real e-commerce users (2021 national college-student data-statistics competition)	KNN, SVM, RF, BPNN, XGBoost	XGBoost best on every metric: accuracy 0.9761, precision 0.9763, recall 0.9761, ROC-AUC 0.9768
[10]	Clickstream sessions from JD.com Mall, China	MBT-POP model vs. baseline classifiers over sliding windows	Best F1 0.9031 at the 2-day window
[11]	Survey data from social-media users	Structural equation modelling	Trust and perceived usefulness drive engagement; privacy concern did not

III. GAP IDENTIFICATION

The literature offers strong individual pieces, namely segmentation methods, explainable prediction and channel-level personalisation. However, no reviewed system combines chronological journey construction, stage and segment classification, sequence-aware prediction and an explainable, marketer-facing Next Best Action layer with a formal no-action option, evaluated on open behavioural data. Table II lists the specific limitations and how this work responds.

IV. PROBLEM FORMULATION

A. Problem Statement

Digital marketing platforms generate large volumes of interaction data (views, searches, cart actions and transactions), yet conventional predictive systems analyse isolated events, such as purchase probability, instead of the customer’s sequential journey [2], [3]. There is a need for an integrated system that converts historical interaction data into chronological customer journeys, classifies journey stage and segment, predicts the likely next action and purchase propensity, and recommends an explainable Next Best Marketing Action, including no action. The target users are Administrators, Marketing Analysts and Marketing Managers.

B. Formal Notation

Let $\mathcal{U}$ be the set of visitors and $\mathcal{I}$ the set of items. Each interaction is a tuple $e = (u, i, c, t)$ with visitor $u \in \mathcal{U}$, item $i \in \mathcal{I}$, event type $c \in \mathcal{C} = \{\text{view, cart, purchase}\}$ and timestamp t . The journey of visitor u is the time-ordered sequence of sessions

$$J_u = (S_{u,1}, S_{u,2}, \dots, S_{u,n_u}), \quad S_{u,k} = (e_1, \dots, e_{m_k}). \quad (1)$$

The framework learns four mappings from J_u observed up to a reference time τ : a stage function $s_u \in \mathcal{S}$, a segment function $g_u \in \mathcal{G}$, a next-event predictor $\hat{e}_u \in \mathcal{C}$, and a propensity estimator $\hat{p}_u \in [0, 1]$, which together determine an action $a_u^* \in \mathcal{A}$.

C. Objectives

The work has six objectives:

- 1) To clean, preprocess and construct chronological, session-aware customer journeys from historical interaction data, with engineered behavioural features such as recency, frequency and view-to-cart ratio.
- 2) To classify customers into journey stages (Initial Engagement, Exploration, Consideration, High Intent, Conversion, Retention, Inactive) and into behavioural, RFM and ML-based segments.
- 3) To predict the next likely customer event and purchase propensity, comparing a baseline (Markov chain), classical ML (Logistic Regression, Random Forest, XGBoost) and a sequence-aware model (LSTM/GRU).
- 4) To design an explainable NBMA engine that recommends actions (cart reminder, email, discount, cross-sell, wait, suppress) with feature-based reasoning.
- 5) To build journey and marketing analytics dashboards (funnel, drop-off, segment and product analytics) usable by digital marketers.
- 6) To compare a generic, a segment-based and a journey-intelligence marketing strategy through offline simulation.

TABLE II
GAP ANALYSIS OF EXISTING APPROACHES

#	Existing approach	Limitation identified	How this work addresses it
1	RFM and clustering-based segmentation [3]	Segments customers on static behavioural summaries; no explicit chronological journey per customer	Session-aware Customer Journey Builder feeding segmentation, not only static RFM scores
2	Big-data strategic marketing and dynamic-capabilities models [4]	Operates at firm or strategy level; lacks a concrete per-customer decision engine	Translates strategy-level capability into a per-customer, rule-based NBMA engine
3	SHAP/QFD explainable CRM-to-CJM framework [6]	Explains engagement and purchase drivers but stops at analysis; no formal suppress or wait decision	Extends explainable prediction into an action layer that models Wait and Suppress Marketing as valid actions
4	Counterfactual dynamic customer-behaviour segmentation [7]	Improves personalisation granularity but not evaluated against alternative marketing decision strategies	Offline strategy-comparison experiment: generic vs. segment-based vs. journey-intelligence (RQ4)
5	Purchase prediction on click-stream and XGBoost data [9], [10]	Optimise prediction accuracy only; stop short of a marketer-facing recommendation	Extends prediction into an explainable NBMA layer with feature-based reasoning

D. Research Questions

- **RQ1:** Does sequence modelling improve next-event prediction over baseline and classical models?
- **RQ2:** Does behavioural data yield separable and interpretable customer segments?
- **RQ3:** Do journey-aware features improve purchase-propensity prediction over static or basic features?
- **RQ4:** Do journey-aware decision rules reduce irrelevant marketing actions compared with generic and segment-based strategies?

V. SCOPE OF THE STUDY

The project is positioned within eight areas of digital marketing. Table III states, for each, how it is covered and what is deliberately left out because the dataset does not support it. This keeps the claims of the paper honest and reproducible.

VI. DATASET AND PREPROCESSING

A. RetailRocket Dataset

We use the RetailRocket e-commerce behavioural dataset [12], an open dataset of visitor interactions with an on-line retail site. Interactions are recorded as view, add-to-cart and transaction events with visitor identifier, item identifier and timestamp. The release also contains item-property and category-tree files that allow items to be grouped by category. The dataset captures on-site behaviour only; it has no search-engine, keyword, social-channel or live campaign-outcome data. For this reason SEO and social-media analytics are treated conceptually (Table III) and all experiments are offline.

B. Event Standardisation and Cleaning

Raw records are validated for missing identifiers and timestamps, converted to a common schema, and mapped to the standard event types $\mathcal{C}$. Exact duplicate records are removed, timestamps are converted to a single time zone, and visitors with implausibly high event rates (possible automated traffic) are flagged and inspected before being excluded. Table IV

reports the raw dataset statistics before cleaning; the counts after cleaning and sessionisation are produced by the pipeline and reported alongside the experimental results.

C. Sessionisation

For each visitor, events are sorted by timestamp. A new session starts whenever the gap between two consecutive events exceeds an inactivity threshold $\theta = 30$ minutes:

$$\text{new session at } e_{j+1} \iff t_{j+1} - t_j > \theta. \quad (2)$$

Session-level attributes (start time, duration, number of events, number of distinct items and categories, and whether it contains a cart or purchase event) are stored for use as features and for journey visualisation.

VII. PROPOSED METHODOLOGY

Fig. 1 gives an end-to-end overview. Raw interaction data are standardised and sessionised; the Customer Journey Builder assembles profiles, stages and features; the ML engine performs segmentation and prediction; the Decision Engine and Personalisation Engine produce an explained NBMA.

A. Customer Journey Builder and Feature Engineering

The Journey Builder assembles each visitor's ordered sessions J_u (Eq. 1) into a Customer Profile, a Journey Stage and a feature vector $\mathbf{x}_u(\tau)$. All features for a given reference time τ use only events with $t \leq \tau$. Table V lists the planned feature groups. Recency, frequency and a monetary proxy are

$$R_u = \tau - t_u^{\text{last}}, \quad F_u = n_u, \quad M_u = \sum_{e \in J_u} \mathbb{1}\{c = \text{purchase}\}, \quad (3)$$

where the monetary component M_u is proxied by the number of purchased items because reliable price information may not be available. The view-to-cart and cart-to-purchase ratios are

$$\rho_u^{vc} = \frac{N_u^{\text{cart}}}{N_u^{\text{view}} + 1}, \quad \rho_u^{cp} = \frac{N_u^{\text{purchase}}}{N_u^{\text{cart}} + 1}, \quad (4)$$

with N_u^c the number of events of type c and the +1 term avoiding division by zero.

TABLE III
SCOPE OF THE STUDY ACROSS DIGITAL-MARKETING AREAS

#	Area	Coverage in this work	Status	Boundary / reason
1	Market and competitor analysis	Positioning against commercial NBA/CDP tools and recent AI-in-marketing literature [1]; gap analysis (Section III)	Conceptual	No live competitor-tracking module
2	Customer and persona analysis	Rule-based, RFM and K-Means segmentation [3]	Implemented	Based on behavioural data only
3	Website and behavioural analysis	Event standardisation, sessions, journey timelines, Customer 360° profiles	Implemented	RetailRocket on-site events [12]
4	SEO and keyword re-search	Reviewed as a complementary acquisition channel	Conceptual	Dataset has no search-engine or keyword data
5	Marketing strategy design	Retargeting, cart reminder, re-engagement, personalised email/channel/timing, cross-sell via the NBMA engine, informed by [4]	Implemented	Rule-based, offline
6	Social media and campaign analytics	Literature on AI-driven personalisation in social engagement [11]	Conceptual	No social-channel or live campaign data
7	AI integration	Baseline, classical and sequence models; K-Means; collaborative filtering; SHAP/rule explanations	Implemented	Offline evaluation
8	Web analytics and conversion/ROI analysis	Executive dashboard, funnel, journey flow, drop-off, segment and product analytics; offline ROI-style evaluation	Implemented (offline)	No live-campaign ROI measurement

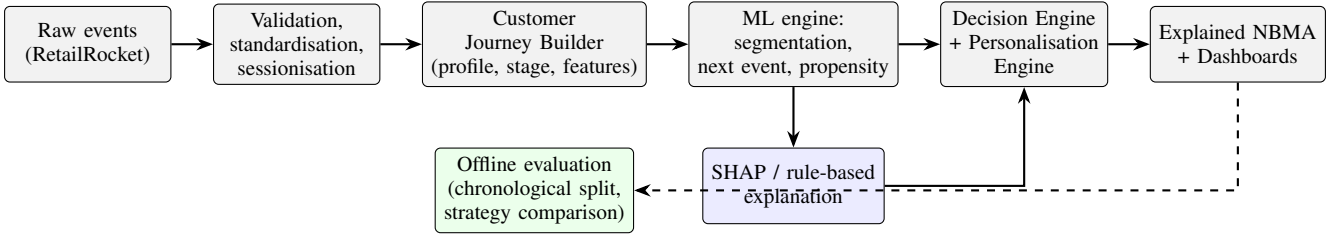


Fig. 1. End-to-end Customer Journey Intelligence pipeline.

TABLE IV
RETAILROCKET DATASET STATISTICS (RAW, AS PUBLISHED ON KAGGLE)

Statistic	Value
Total events	2,756,101
Unique visitors	1,407,580
Unique items (in events)	235,061
Observation period	May–Sep 2015
View events	2,664,312
Add-to-cart events	69,332
Transaction events	22,457
Visitors with ≥ 1 purchase	11,719

B. Journey-Stage Classification

Each visitor is assigned one of seven stages $\mathcal{S}$: Initial Engagement, Exploration, Consideration, High Intent, Conversion, Retention or Inactive. Stage assignment uses transparent,

TABLE V
PLANNED BEHAVIOURAL FEATURE GROUPS

Group	Features
RFM	Recency, frequency (sessions), monetary proxy
Funnel	View-to-cart ratio, cart-to-purchase ratio, abandoned-cart flag
Session	Mean and last session length, events per session, inter-session gap
Catalogue	Distinct items and categories viewed, repeat-view count
Temporal	Hour-of-day and day-of-week of last activity, days since last purchase
Sequence	Last L event types (for sequence models)

ordered rules over the features so that marketers can audit each label; Table VI shows the initial rule set. Thresholds are

TABLE VI
INITIAL JOURNEY-STAGE RULES (THRESHOLDS CONFIGURABLE)

Stage	Illustrative condition
Initial Engage-ment	First or very few sessions; views only
Exploration	Multiple views across items or categories, no cart event
Consideration	Repeated views of the same items; cart activity beginning
High Intent	Recent add-to-cart with no subsequent purchase
Conversion	Purchase within the current session
Retention	Purchase in the past and recent return activity
Inactive	No activity beyond the inactivity window

configurable parameters tuned on the training period only.

C. Customer Segmentation

Segmentation is performed in three complementary ways: (i) rule-based behavioural segments (for example, window shoppers, cart abandoners, repeat buyers); (ii) RFM segments obtained from quantile scores of R_u , F_u and M_u ; and (iii) ML-based segments from K-Means over standardised features. K-Means is minimising the within-cluster sum of squares

$$\min_{\{\mathcal{G}_k\}} \sum_{k=1}^K \sum_{\mathbf{x}_u \in \mathcal{G}_k} \|\mathbf{x}_u - \boldsymbol{\mu}_k\|^2. \quad (5)$$

K-Means with RFM-type features is chosen because it is the most frequently used combination in the literature, not because it has been shown to be best [3]. The number of clusters K is selected using the silhouette score

$$\text{sil}(u) = \frac{b(u) - a(u)}{\max\{a(u), b(u)\}}, \quad (6)$$

where $a(u)$ is the mean intra-cluster distance and $b(u)$ the mean distance to the nearest other cluster, together with a business-interpretability check of the cluster profiles (RQ2).

D. Next-Event Prediction

Next-event prediction estimates $\hat{e}_u = \arg \max_{c \in \mathcal{C}} P(c \mid \text{history})$. Three model families are compared.

Markov-chain baseline. A first-order transition matrix is estimated from training sequences,

$$P(c_{j+1} = b \mid c_j = a) = \frac{N(a \rightarrow b)}{\sum_{b'} N(a \rightarrow b')}. \quad (7)$$

Classical machine learning. Logistic Regression, Random Forest and XGBoost are trained on aggregated features $\mathbf{x}_u(\tau)$. XGBoost is included because gradient-boosted trees have outperformed traditional classifiers for purchase-behaviour prediction [9], and clickstream-based intent modelling supports session-level prediction [10].

Sequence-aware model. An LSTM or GRU consumes the embedded sequence of the last L events (event type, optional

TABLE VII
ILLUSTRATIVE NBMA DECISION RULES (PRIORITY ORDER)

P	Condition	Action
1	Purchase within the last Δ days	Wait / Suppress
2	High Intent, cart present, no purchase	Cart Reminder
3	Consideration, high $\hat{p}_u$	Targeted Discount
4	Retention segment, past purchase	Cross-Sell
5	Inactive, previously engaged	Re-engagement Email
6	Low $\hat{p}_u$, early stage	Wait

item category and time gap) and outputs a softmax distribution over $\mathcal{C}$:

$$\mathbf{h}_j = \text{LSTM}(\mathbf{z}_j, \mathbf{h}_{j-1}), \quad \hat{\mathbf{y}} = \text{softmax}(\mathbf{W}\mathbf{h}_L + \mathbf{b}). \quad (8)$$

The sequence model is optional in the implementation stack and is used to test RQ1.

E. Purchase-Propensity Prediction

Purchase propensity is the probability that a visitor purchases within a horizon H after the reference time τ :

$$\hat{p}_u = P(y_u = 1 \mid \mathbf{x}_u(\tau)), \quad y_u = \mathbb{I}\{\text{purchase in } (\tau, \tau + H]\}. \quad (9)$$

Purchases are rare relative to views, so class imbalance is handled with class weighting (or the `scale_pos_weight` option in XGBoost), and PR-AUC is reported as a supplementary metric. To answer RQ3, each model is trained twice: once with basic static features (for example, total views and total carts) and once with the full journey-aware feature set of Table V.

F. Explainable NBMA Decision Engine

The decision engine maps model outputs to an action in

$$\mathcal{A} = \{\text{Cart Reminder, Email, Discount, Cross-Sell, Wait, Suppress}\}. \quad (10)$$

The policy is a priority-ordered rule set over the stage s_u , segment g_u , predicted next event $\hat{e}_u$, propensity $\hat{p}_u$ and time since last purchase r_u :

$$a_u^* = \pi(s_u, g_u, \hat{e}_u, \hat{p}_u, r_u). \quad (11)$$

Table VII shows the illustrative rules. The suppression rule has the highest priority, so recently converted customers are not contacted unnecessarily. Algorithm 1 summarises the per-customer procedure.

G. Personalisation Engine

The Personalisation Engine adds product, channel and timing suggestions to the chosen action. Product suggestions use item-based collaborative filtering with cosine similarity between item interaction vectors,

$$\text{sim}(i, j) = \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\| \|\mathbf{v}_j\|}, \quad (12)$$

Algorithm 1 Journey Construction and NBMA Recommendation

Require: Event log E , reference time τ , thresholds θ, Δ, λ

Ensure: Action a_u^* and explanation for each visitor u

```

1: Clean and standardise  $E$ ; keep events with  $t \leq \tau$ 
2: for each visitor  $u$  do
3:   Sort events; split into sessions using Eq. (2)
4:   Build profile and features  $\mathbf{x}_u(\tau)$  (Eq. (3), (4))
5:    $s_u \leftarrow$  stage rules;  $g_u \leftarrow$  segment model
6:    $\hat{e}_u \leftarrow$  next-event model;  $\hat{p}_u \leftarrow$  propensity model
7:   if  $r_u \leq \Delta$  then
8:      $a_u^* \leftarrow$  Suppress
9:   else
10:     $a_u^* \leftarrow$  first matching rule in Table VII with threshold
         $\lambda$  on  $\hat{p}_u$ 
11:   end if
12:   Attach explanation (rule trace and top SHAP features)
13: end for

```

where $\mathbf{v}_i$ is the vector of visitor interactions with item i . Channel and timing suggestions (for example, email versus on-site message, and preferred hour of day) are derived from each customer's historical activity pattern. Because the dataset has no channel-response data, these are heuristic suggestions rather than learned channel-response models.

H. Explainability

Each recommendation is accompanied by an explanation in two layers. The rule layer records which rule fired and the values of the features it tested. For the tree-based propensity model, a SHAP layer lists the features that contributed most to $\hat{p}_u$, using Shapley values

$$\phi_f = \sum_{Q \subseteq F \setminus \{f\}} \frac{|Q|! (|F| - |Q| - 1)!}{|F|!} [v(Q \cup \{f\}) - v(Q)], \quad (13)$$

where F is the feature set and $v(\cdot)$ the model output for a feature subset. This follows the pattern of integrating Shapley explanations with an operational decision layer [6] and addresses the finding that trust and perceived usefulness drive engagement with AI-driven personalisation [11].

VIII. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Layered Architecture

Fig. 2 shows the complete layered architecture, which has nine layers and sixteen numbered core components. The *data source and storage layer* holds the RetailRocket files and a central PostgreSQL store with raw, processed and results tables. The *data pipeline layer* ingests and validates records (schema, null, duplicate and timestamp checks), standardises events to common types and assigns session identifiers with the 30-minute inactivity rule. The *journey processing layer* builds the per-customer chronological timeline, engineers features and assigns the journey stage. The *machine learning layer* performs segmentation, next-event prediction and propensity

TABLE VIII
TOOLS AND TECHNOLOGIES

Category	Tools / technologies
Frontend	React (or Next.js), TypeScript, Tailwind CSS, shadcn/ui, Recharts
Backend	Python, FastAPI
Database	PostgreSQL (Supabase / Neon)
Machine learning	Pandas, NumPy, Scikit-learn, XGBoost/LightGBM; optional PyTorch/TensorFlow (LSTM/GRU); SHAP
Visualisation	Recharts, Plotly
Deployment	Vercel (frontend), Render/Railway (backend), Supabase/Neon (database)
Dataset	RetailRocket e-commerce behavioural dataset [12]

prediction, and the *decision and intelligence layer* turns these outputs into an NBMA with personalisation and an explanation. The *evaluation and experiment layer* is strictly offline: it measures models for RQ1–RQ3 and simulates the three marketing strategies for RQ4, and saves its results to PostgreSQL. The *application layer* exposes FastAPI services and role-based access control, the *presentation layer* renders the dashboards, and the *user layer* serves Administrators, Marketing Analysts and Marketing Managers.

B. Technology Stack

Table VIII lists the technology stack, which implements the layers of Fig. 2. The backend exposes the journey, prediction and recommendation services through a REST API, and the frontend consumes it to render the dashboards.

C. Marketer-Facing Dashboards

The application provides the following views: (i) an executive dashboard with headline engagement and conversion indicators; (ii) a conversion funnel (view, cart, purchase) with stage-wise drop-off; (iii) a journey-flow view showing transitions between journey stages; (iv) segment analytics describing the size and behaviour of each segment; (v) product analytics, including most viewed, most carted and most purchased items; and (vi) a Customer 360° profile showing the chronological journey, current stage and segment, predicted next event, propensity, recommended NBMA and its explanation. Admins manage data and model runs, while Marketing Analysts and Managers use the analytical and recommendation views.

D. Development Phases

The development proceeds through eight phases, summarised in Table IX.

IX. EXPERIMENTAL DESIGN

A. Data Splitting and Leakage Control

All models are trained on an earlier period and tested on a later period using chronological splits, never random splits. A validation window immediately before the test window is used for tuning. For any customer, features are computed only from

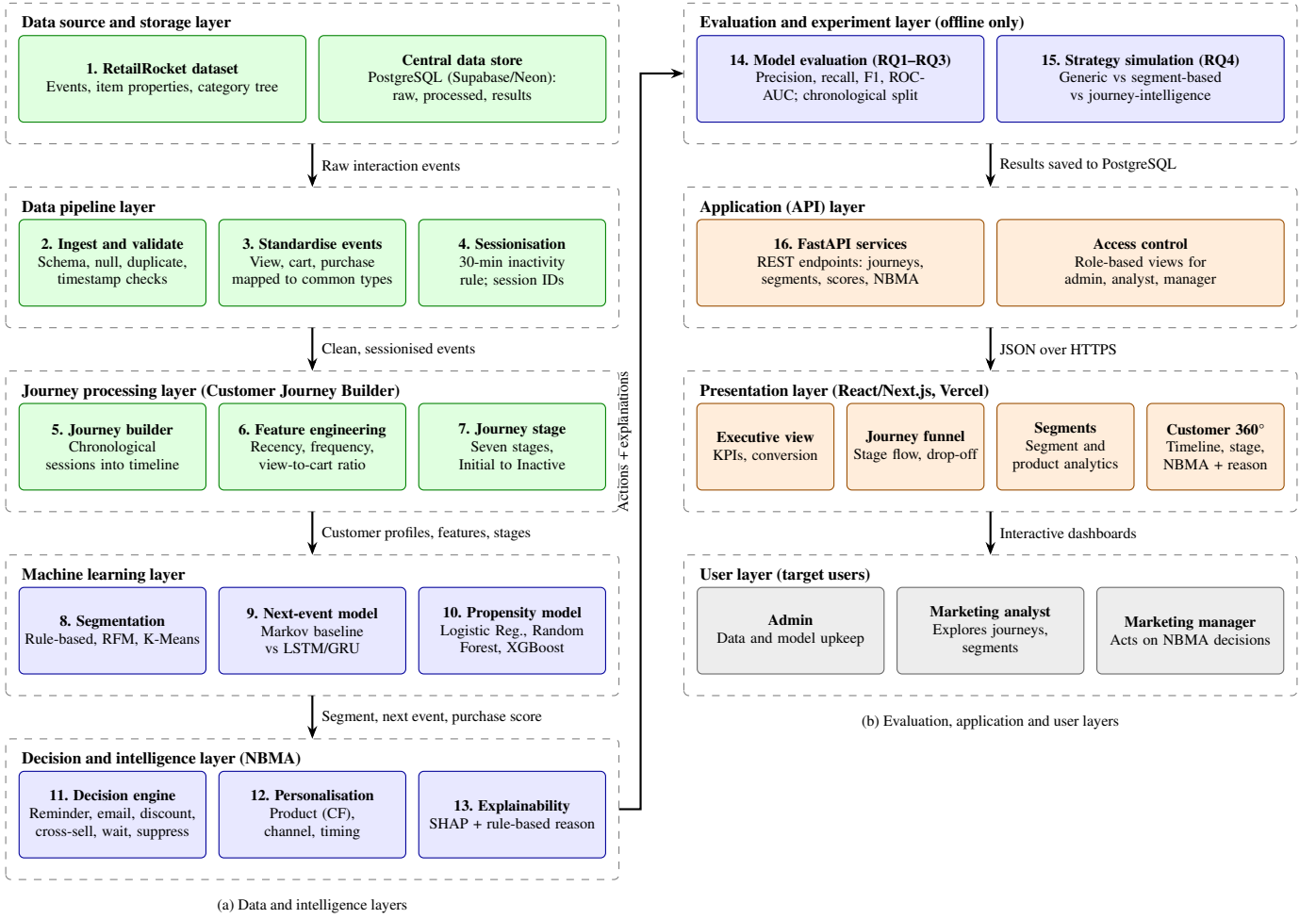


Fig. 2. Layered system architecture of the Customer Journey Intelligence platform. Green: data layers; blue: intelligence layers; orange: application layers; grey: users. Numbers 1–16 denote the core components.

TABLE IX
DEVELOPMENT PHASES

#	Phase and main activity
1	Research and dataset finalisation: literature review, gap analysis, dataset selection
2	Data pipeline: validation, standardisation, sessionisation, storage
3	Core backend: journey builder, profile and feature services, API
4	ML modelling: segmentation, next-event and propensity models, comparison
5	Decision-engine integration: NBMA rules, personalisation, explanations
6	Frontend dashboards: funnel, journey, segment, product, Customer 360°
7	Strategy-comparison experiments: offline simulation of three strategies
8	Documentation: report, paper and reproducibility package

TABLE X
MODELS AND PLANNED HYPERPARAMETER SEARCH

Model	Planned search space
Markov chain	First-order transitions; smoothing constant
Logistic Regression	Regularisation strength C ; class weights
Random Forest	Number of trees, maximum depth, minimum leaf size
XGBoost	Learning rate, tree depth, number of rounds, <code>scale_pos_weight</code>
LSTM/GRU	Hidden size, layers, dropout, sequence length L

after τ . Stage and segment thresholds and cluster centroids are fitted on the training period only.

B. Models and Tuning

Table X lists the model families and the planned hyperparameter search. Search spaces are fixed before running the experiments and are tuned on the validation window.

events with $t \leq \tau$, and the label window $(\tau, \tau + H]$ is strictly

TABLE XI
RESEARCH QUESTIONS AND EVALUATION PLAN

RQ	Comparison	Evidence
RQ1	Markov vs. classical vs. LSTM/GRU on next-event prediction	Macro precision, recall, F1, accuracy
RQ2	K-Means vs. RFM vs. rule-based segments	Silhouette score; cluster profiles
RQ3	Basic vs. journey-aware features on propensity	F1, ROC-AUC, PR-AUC; SHAP
RQ4	Generic vs. segment vs. journey strategy	$Prec_{target}$, Cov, Unn

C. Evaluation Metrics

Classification models are evaluated using accuracy, precision, recall, F1-score and ROC-AUC, with macro-averaging for multi-class next-event prediction:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}. \quad (14)$$

Because purchases are rare, accuracy alone can be misleading, and PR-AUC is reported as a supplementary measure. Segmentation quality is assessed with the silhouette score (Eq. 6) and the interpretability of cluster profiles.

D. Offline Strategy-Comparison Experiment

Three strategies are simulated on the held-out period: (i) a *generic* strategy that applies the same action to all customers, (ii) a *segment-based* strategy that assigns an action per segment, and (iii) the *journey-intelligence* strategy of Section VII-F. Let $\mathcal{T}$ be the set of customers receiving an intervention (any action other than Wait or Suppress), $\mathcal{B}$ the set of customers who purchase within the evaluation horizon, and $\mathcal{Z}$ the set of customers who purchased within the last Δ days before τ . The offline proxies are

$$Prec_{target} = \frac{|\mathcal{T} \cap \mathcal{B}|}{|\mathcal{T}|}, \quad Cov = \frac{|\mathcal{T} \cap \mathcal{B}|}{|\mathcal{B}|}, \quad Unn = \frac{|\mathcal{T} \cap \mathcal{Z}|}{|\mathcal{T}|}. \quad (15)$$

The action distribution of each strategy is also reported. These proxies describe how well a strategy aligns interventions with observed purchasing; because the dataset contains no treatment-outcome data, they do not estimate causal uplift.

E. Mapping of Research Questions to Experiments

Table XI links each research question to its experiment and evidence.

F. Reproducibility

All preprocessing, model training and evaluation scripts will be released with fixed random seeds, the exact chronological cut-off dates, the final threshold values and the environment specification, so that all reported tables can be regenerated from the open dataset.

TABLE XII
MODEL COMPARISON FOR NEXT-EVENT AND PURCHASE PREDICTION (%). **ILLUSTRATIVE DEMONSTRATION VALUES (RED), NOT MEASURED RESULTS; REPLACE WITH EXPERIMENTAL OUTPUT.**

Model	Acc.	Prec.	Rec.	F1	AUC
Markov chain	71.0	64.0	66.0	65.0	69.0
Logistic Regression	78.0	72.0	70.0	71.0	80.0
Random Forest	84.0	79.0	77.0	78.0	87.0
XGBoost	87.0	83.0	81.0	82.0	91.0
LSTM/GRU	89.0	85.0	84.0	84.0	92.0

TABLE XIII
OFFLINE STRATEGY COMPARISON (%). **ILLUSTRATIVE DEMONSTRATION VALUES (RED), NOT MEASURED RESULTS; REPLACE WITH EXPERIMENTAL OUTPUT.**

Strategy	$Prec_{target}$	Cov	Unn
Generic	3.0	100.0	15.0
Segment-based	7.0	78.0	12.0
Journey-intelligence	15.0	64.0	2.0

X. RESULTS AND DISCUSSION

The results will be reported as three groups of outputs: the journey-stage and funnel analysis; the model-comparison table (Table XII) with SHAP-based explanations; and the offline strategy-comparison table (Table XIII). The numerical entries shown in red in both tables are illustrative demonstration values included only to show the intended reporting format; they are not measured results and will be replaced with the outputs of the RetailRocket experiments. We do not quote figures from other papers as targets, since those studies use different datasets and settings [9], [10]. All results will be presented as offline, simulated evaluation.

A. Discussion Plan by Research Question

RQ1. We will test whether the sequence-aware model improves macro-F1 over the Markov baseline and the classical models. If it does not, we will report that, since on short sessions aggregated features may already capture most of the signal.

RQ2. We will examine whether K-Means on behavioural features gives well-separated clusters (high silhouette) with distinct, nameable profiles, in line with the segmentation literature [3].

RQ3. We will compare propensity models with and without journey-aware features and use SHAP to show which journey features matter, consistent with findings that modelling behaviour transitions improves personalisation granularity [7].

RQ4. We will test whether the journey-intelligence strategy reduces unnecessary interventions (lower Unn) while keeping coverage and targeting precision competitive with the other strategies. A reduction in interventions inevitably lowers coverage, so the three metrics will be interpreted together rather than separately.

XI. ETHICS, PRIVACY AND SOCIETAL IMPACT

The RetailRocket data are anonymised, and the framework works with behavioural identifiers rather than personal attributes. In a production setting, behavioural profiling should follow applicable data-protection rules, including purpose limitation, consent where required and data minimisation. The explainable design supports this: each recommendation can be traced to specific behaviours, and the Wait and Suppress actions limit unnecessary contact, reducing message fatigue. Because trust is a driver of engagement with AI personalisation [11], transparency about why a customer is contacted is also commercially sensible. The framework contributes to SDG 8 by helping organisations use marketing resources more efficiently, and to SDG 9 by providing an open, deployable analytics infrastructure.

XII. LIMITATIONS AND THREATS TO VALIDITY

Offline evaluation. The dataset contains no treatment-outcome data, so strategy comparison is a simulation of alignment with observed purchases and does not measure causal uplift. A customer who would have purchased anyway counts as a successful target, so the proxies may overstate the value of any strategy. Live A/B testing is needed before deployment.

Dataset scope. RetailRocket covers a single retailer and a limited period, and has no search, social, channel-response or price-rich campaign data. Findings may not generalise to other domains. SEO, social-media analytics and competitor tracking are therefore treated conceptually.

Rule-based policy. The NBMA thresholds are design parameters tuned on the training period; they are transparent but not learned from interventions. A reinforcement-learning or uplift-modelling policy could be explored with experimental data.

Cold start and sparse histories. Many visitors have very short journeys. Their stage and propensity estimates are less reliable, and the framework defaults to Wait for low-evidence customers.

Comparability with prior work. Metrics reported in the literature come from different datasets and tasks, so they are used for context only and not as direct benchmarks [9], [10].

XIII. CONCLUSION AND FUTURE WORK

This paper presented an explainable Customer Journey Intelligence framework that models customer behaviour as a sequence, classifies journey stage and segment, predicts next actions and purchase propensity, and recommends a Next Best Marketing Action that includes a deliberate no-action option. The framework spans eight digital-marketing capability areas, is delivered through marketer-facing dashboards, and is evaluated against four explicit research questions using chronological splits and an offline strategy comparison on open data. The design aims to show how sequential behavioural intelligence can make digital marketing more personalised, explainable and efficient within an academically honest and reproducible framework.

Future work includes: completing and reporting the experiments described here; validating the policy through live A/B tests; replacing hand-set thresholds with uplift modelling or reinforcement learning; incorporating search, social and channel-response data to bring SEO and social analytics into scope; and evaluating the framework on additional retail datasets.

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